

Curriculum Vitae: Luke P. Corcos

Email: lucorcos@gmail.com

1-(410)-430-0185

Education

University of California, Berkeley, CA Ph.D., Applied Mathematics Advisor: Professor James A. Sethian Thesis title: <i>A Hybrid Finite Difference Level Set–Implicit Mesh Discontinuous Galerkin Method for Multi-Layer Coating Flow Problems</i>	2016-2023
University of Maryland, College Park, MD Bachelor of Science, Mathematics and Physics High Honors in Physics	2012-2016

Positions

Lawrence Berkeley National Laboratory, <i>Graduate Student Researcher</i>	2017-2023
University of California, Berkeley, <i>Graduate Student Instructor</i>	2017
Joint Quantum Institute, <i>Undergraduate Researcher</i>	2014-2016

Publications

- L. P. Corcos, R. I. Saye, and J. A. Sethian, *A hybrid finite difference level set–implicit mesh discontinuous Galerkin method for multi-layer coating flows*, Journal of Computational Physics, 507, 112960 (2024) [doi:10.1016/j.jcp.2024.112960](https://doi.org/10.1016/j.jcp.2024.112960).
- L. P. Corcos, *A Hybrid Finite Difference Level Set–Implicit Mesh Discontinuous Galerkin Method for Multi-Layer Coating Flow Problems*, Ph.D. thesis, University of California Berkeley, 2023. [Link](#).
- A. D. Cimmarusti, Z. Yan, B. D. Patterson, L. P. Corcos, L. A. Orozco, and S. Deffner, *Environment-Assisted Speed-up of the Field Evolution in Cavity Quantum Electrodynamics*, Phys. Rev. Lett. 114, 233602 (2015) [doi:10.1103/PhysRevLett.114.233602](https://doi.org/10.1103/PhysRevLett.114.233602).

Honors and Awards

Benard Friedman Memorial Prize in Applied Mathematics, UC Berkeley	2024
High Honors in Physics, UMD	2016

University Honors, UMD
Banneker Key Scholarship, UMD

2016
2012-2016

Research Grants

- High Performance Computing for Manufacturing, US Dept. of Energy, *Modeling Coating Flow and Dynamics during Drying*, with PPG Industries, Inc. (2020).

Research Interests

High-order numerical methods for multi-physics problems, interfacial fluid dynamics, phase change, thermodynamics, level set methods, discontinuous Galerkin (DG) methods, PDEs, numerical linear algebra, high-performance computing.